

Day and Night: 24-Hour Control Room

I can use a rotating-Earth model to collect and explain evidence across one complete day-night cycle.

Four quarter turns

use evidence

A complete evidence record

use evidence

An explanation tested against a misconception

[Open the original interactive lesson](#)

What success looks like

- Record four clear positions for one place on Earth.
- Link each position to light, darkness and the observer's view.
- Use the evidence to show how rotation explains the observed day-night pattern without changing the lamp.

Retrieve and reconnect

- Last lesson: what moved in our model and what stayed fixed?
- Last lesson: why does the Sun appear to cross the sky?
- Last week: how did observer position matter in the Moon model?
- Lead-in: what evidence would prove that a whole cycle was observed?

A pupil says, 'Clouds make night because they hide the Sun.' Which first response would you test?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Agree: thick clouds create night.
- B. Track a marked place as Earth turns, with the lamp kept on.
- C. Use a brighter lamp so night disappears.

Look closely · what is the evidence?

Observation 1: the lamp is on and the marker is at the light-dark boundary.

Observation 2: the lamp is still on and the marker is fully lit.

Observation 3: the lamp remains on and the marker is unlit.

Look closely · what is the evidence?



Marker at the boundary, lamp on



Marker fully lit, lamp on



Marker unlit, lamp still on

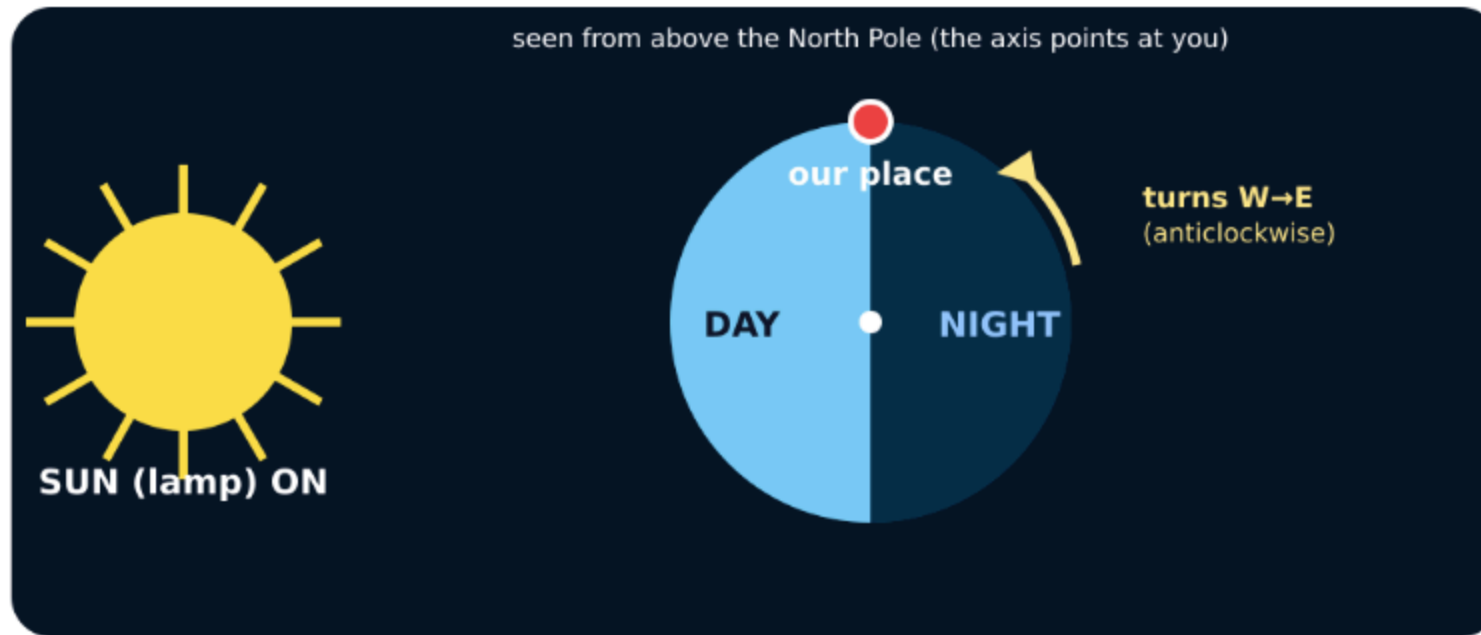
Words we will use

- cycle — repeating sequence
- rotation — one full turn
- evidence — what we observed

Watch the model

- Place one marker on the globe and start it at sunrise, where it crosses into the lit half.
- Turn the globe west to east in equal quarter turns. Pause and record whether the marker is lit and which way it faces.
- Return the marker to the start. Four ordered observations now represent one simplified 24-hour cycle.

Watch the model



Rotating Earth model with lamp, cloud test and lamp test

The lamp stayed on throughout, but the marker became dark. Which evidence explains this?

- A. The marker rotated onto the side facing away from the lamp.
- B. The lamp stopped producing light for half the turn.
- C. Clouds formed on the globe during the turn.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Add an observer arrow at each marker position to show which way the local person faces.
- Describe the Sun's apparent east-to-west position without claiming that the lamp moved; west-to-east Earth rotation creates that observed direction.
- Link all four rows: because Earth rotated, the observer experienced sunrise, day, sunset and night.

Find and repair the misconception

Model glitch: clouds make night by covering the Sun.

A. Keep it: darkness always means cloud cover.

B. Repair it: rotation moves a place onto Earth's unlit side.

C. Repair it: the Sun switches off each night.

24-hour evidence console

Commit each control-room observation to the correct point in the cycle.

- Select one observation.
- Place it at the matching cycle point.
- Repair any mismatch using lit or unlit evidence.
- Freeze the record and state the repeating pattern.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- Sunrise
- Middle of the day
- Sunset
- Middle of the night

Use two frozen rows to show how rotation explains night while the lamp remains on and no cloud is present.

Your independent mission

Run or represent one complete four-position rotation and produce a control-room evidence record.

Record: Four ordered rows recording marker position, light state, apparent Sun position and one final because conclusion.

Choose your support and challenge

- Supported: Complete four rows using picture choices for sunrise, day, sunset and night.
- Standard: Record four quarter turns and explain the complete cycle using rotation and apparent movement.
- Stretch: Use the four rows to show how west-to-east rotation explains the pattern. Compare this explanation with the cloud and daily-Sun-orbit ideas without claiming the model alone decides between every possible sky model.

Show what you know

Use two evidence rows to explain one full day-night cycle.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.